

CSE-719(Distributed and Cloud Computing)

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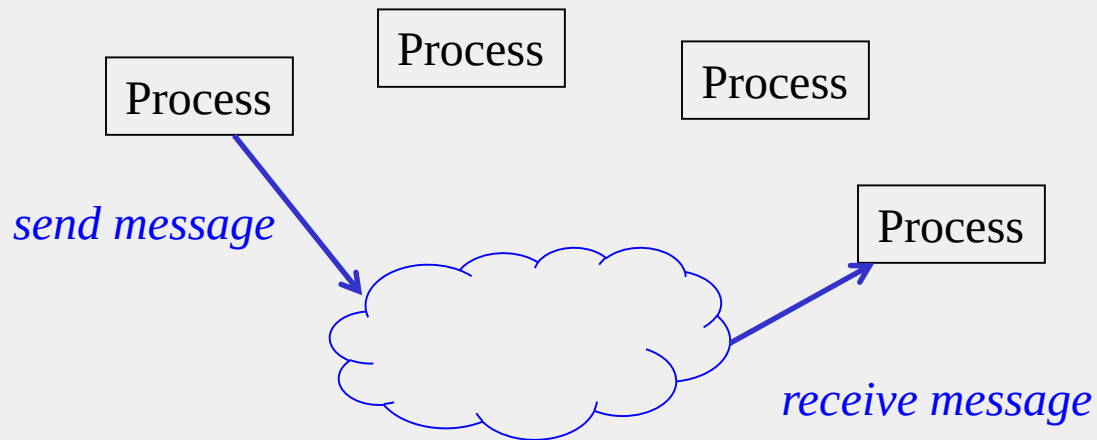
Department of Computer Science and Engineering

University of Chittagong

Lecture 9: Distributed Shared Memory

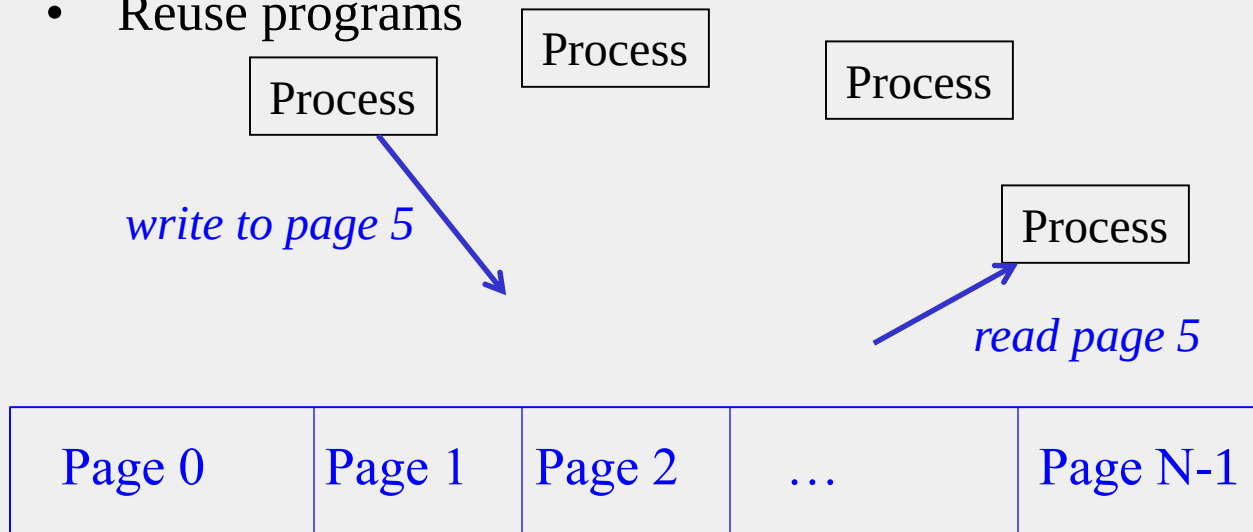
So Far ...

- Message passing network



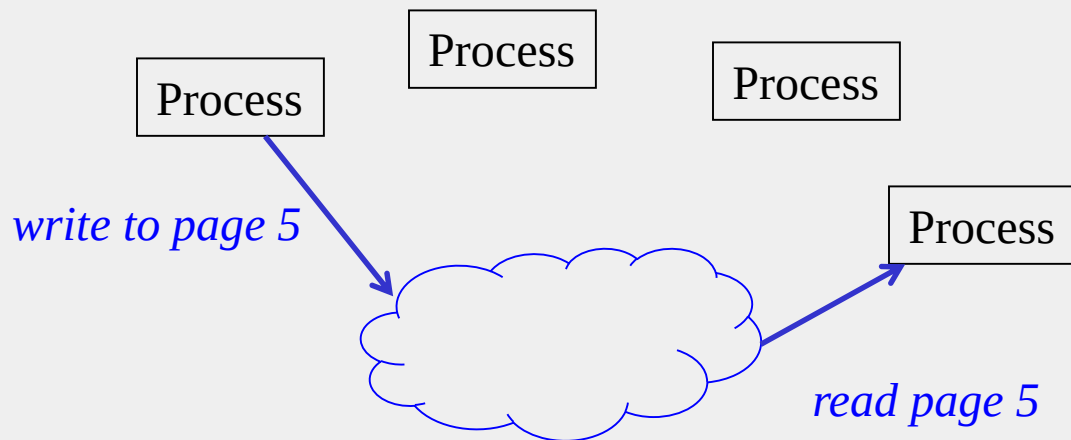
But what if ...

- Processes could *share* memory pages instead?
- Makes it convenient to write programs
- Reuse programs



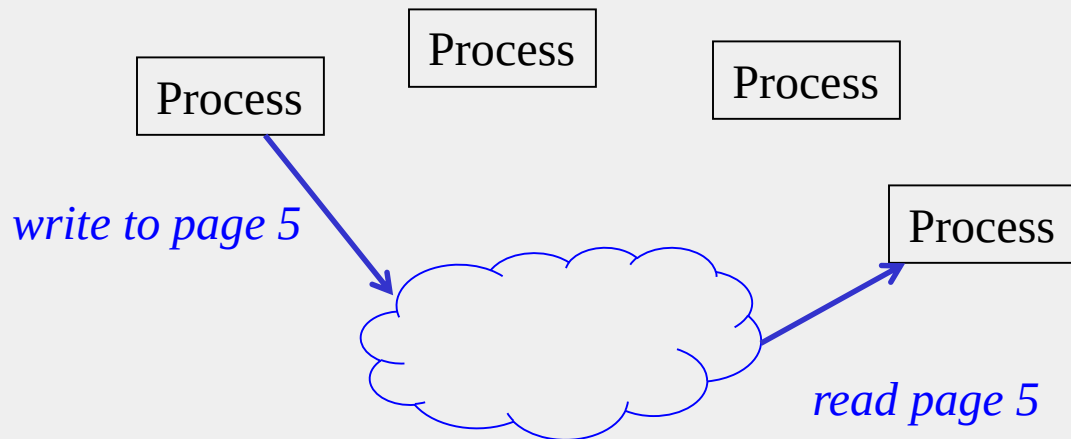
Distributed Shared Memory

- Distributed Shared Memory = processes virtually share pages
- How do you implement DSM over a message-passing network?



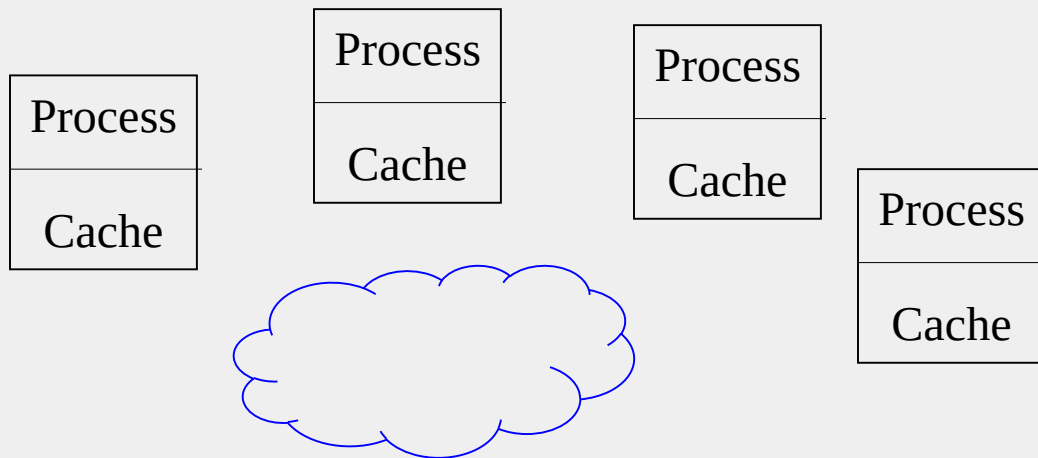
In fact ...

1. Message-passing can be implemented over DSM!
 - Use a common page as buffer to read/write messages
2. DSM can be implemented over a message-passing network!



DSM over Message-Passing Network

- **Cache** maintained at each process
 - Cache stores pages accessed recently by that process
- Read/write first goes to cache



DSM over Message-Passing Network (2)

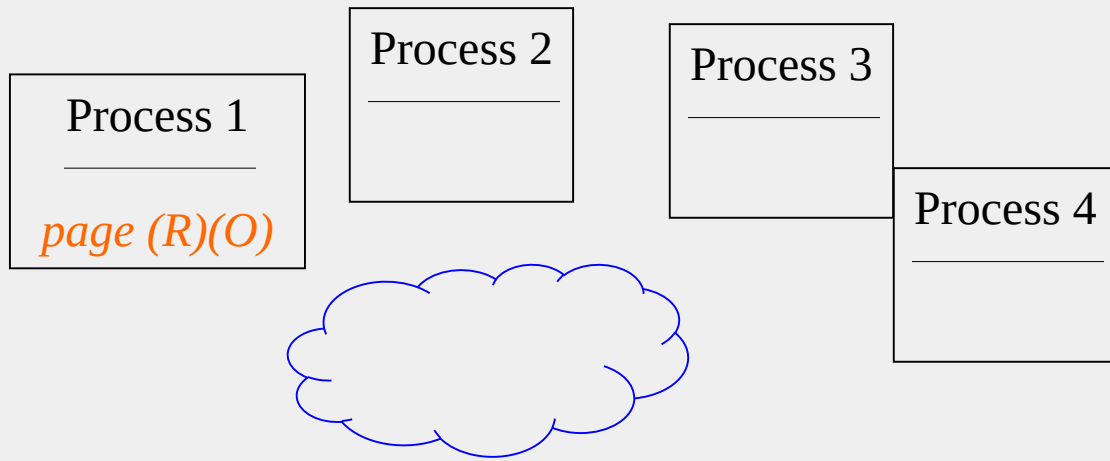
- Pages can be mapped in local memory
- When page is present in memory, page hit
- Otherwise, *page fault* (kernel trap) occurs
 - Kernel trap handler: invokes the DSM software
 - May contact other processes in DSM group, via multicast

DSM: Invalidate Protocol

- Owner = Process with latest version of page
- Each page is in either R or W state
- When page in R state, owner has an R copy, but other processes may also have R copies
 - but no W copies exist
- When page is in W state, only owner has a copy

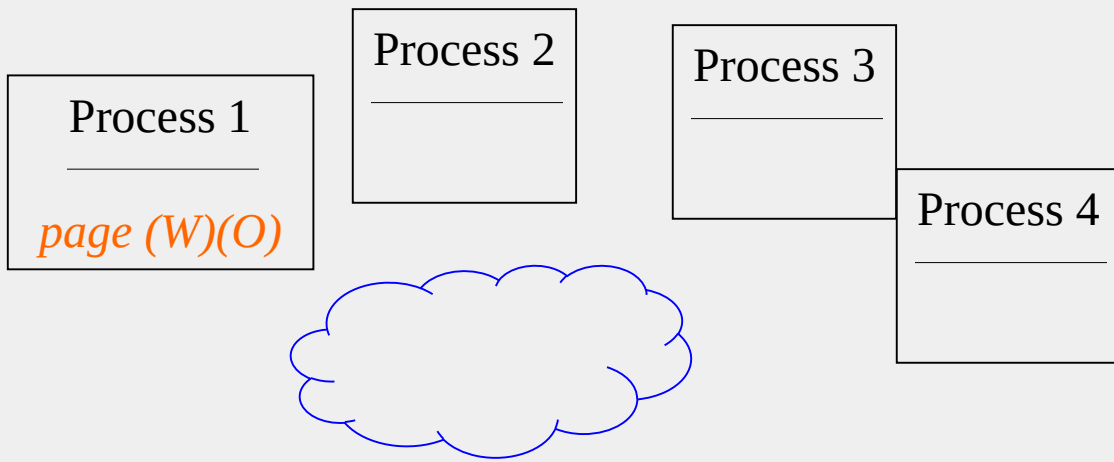
Process 1 Attempting a Read: Scenario 1

- Process 1 is owner (*O*) and has page in R state
- *Read from cache. No messages sent.*



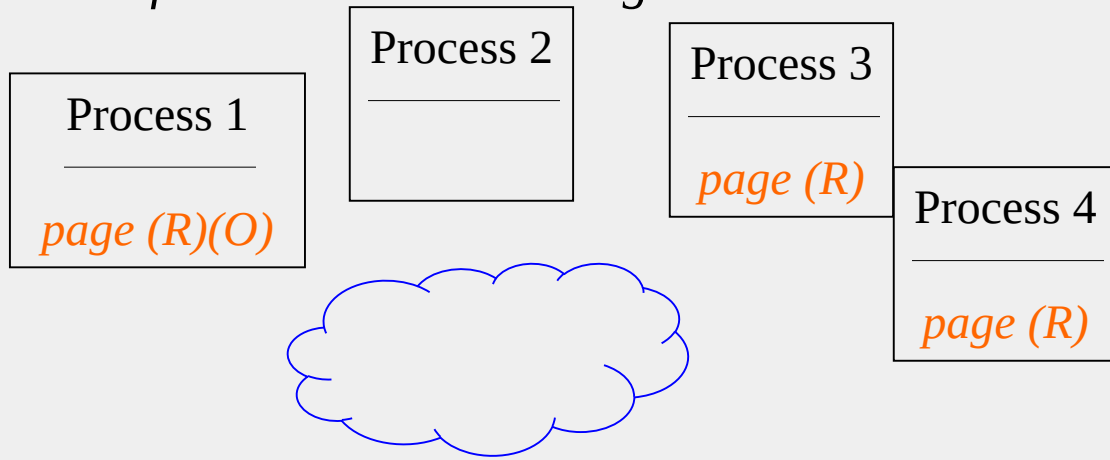
Process 1 Attempting a Read: Scenario 2

- Process 1 is owner (*O*) and has page in *W* state
- *Read from cache. No messages sent.*



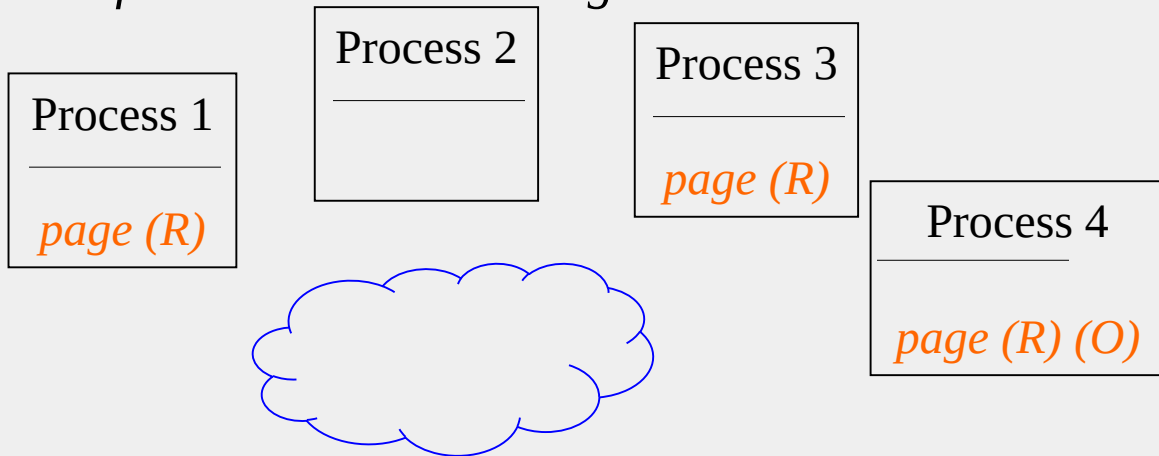
Process 1 Attempting a Read: Scenario 3

- Process 1 is owner (*O*) and has page in R state
- Other processes also have page in R state
- *Read from cache. No messages sent.*



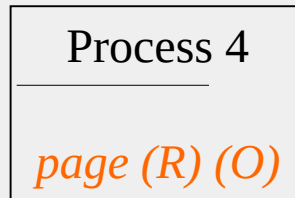
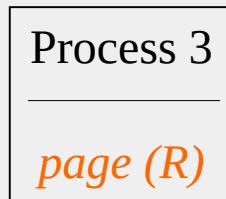
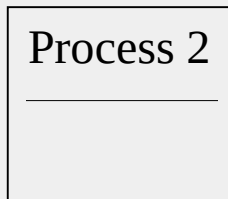
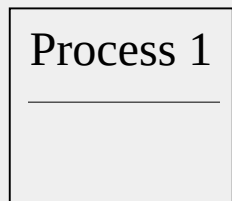
Process 1 Attempting a Read: Scenario 4

- Process 1 has page in R state
- Other processes also have page in R state, and someone else is owner
- *Read from cache. No messages sent.*



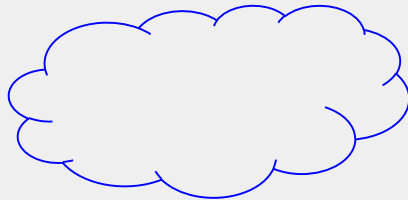
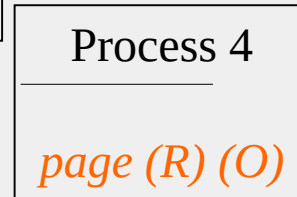
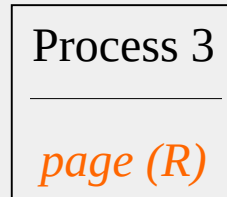
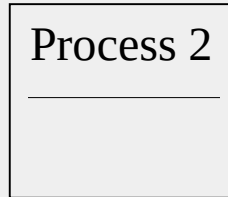
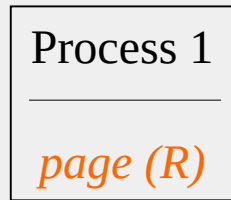
Process 1 Attempting a Read: Scenario 5

- Process 1 does not have page
- Other process(es) has/have page in (R) state
- *Ask for a copy of page. Use multicast.*
- *Mark it as R*
- *Do Read*



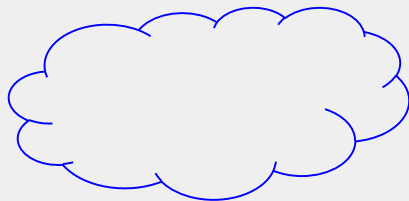
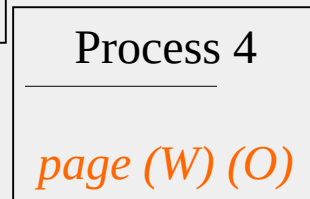
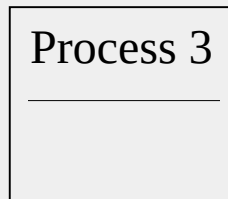
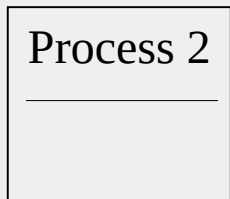
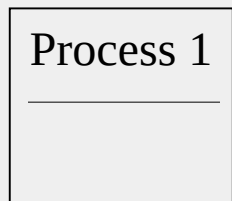
End State: Read Scenario 5

- Process 1 does not have page
- Other process(es) has/have page in (R) state
- *Ask for a copy of page. Use multicast.*
- *Mark it as R*
- *Do Read*



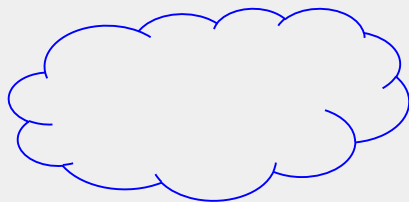
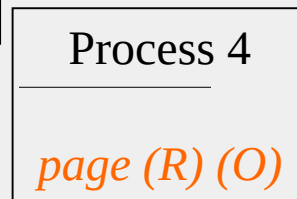
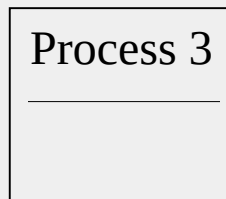
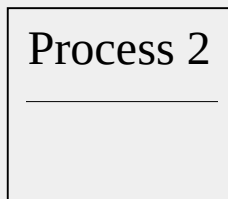
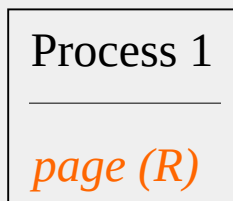
Process 1 Attempting a Read: Scenario 6

- Process 1 does not have page
- Another process has page in (W) state
- *Ask other process to degrade its copy to (R). Locate process via multicast*
- *Get page; mark it as R*
- *Do Read*



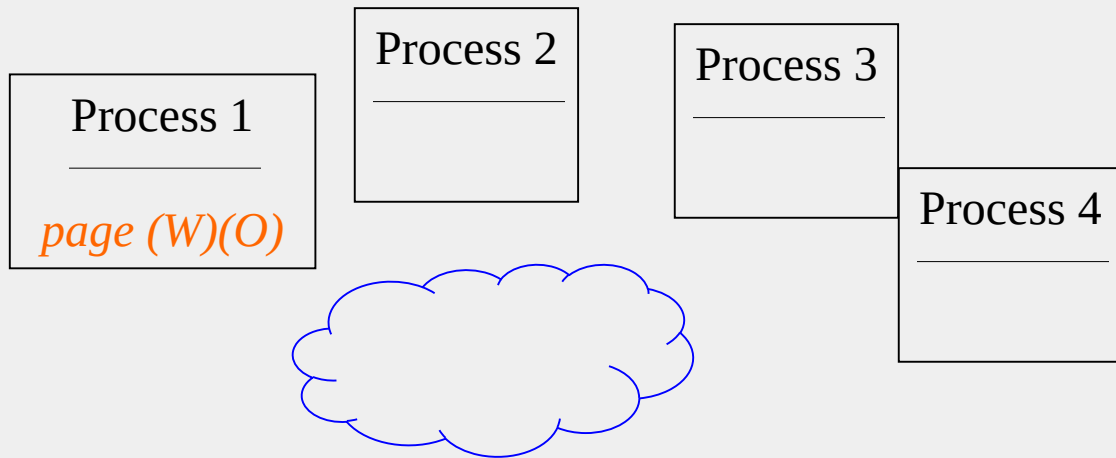
End State: Read Scenario 6

- Process 1 does not have page
- Another process has page in (W) state
- *Ask other process to degrade its copy to (R). Locate process via multicast*
- *Get page; mark it as R*
- *Do Read*



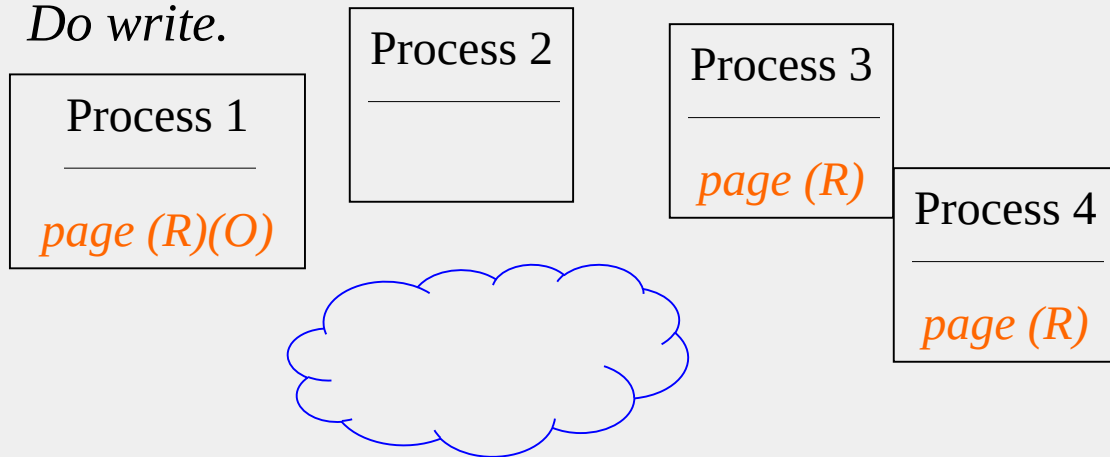
Process 1 Attempting a Write: Scenario 1

- Process 1 is owner (*O*) and has page in *W* state
- *Write to cache. No messages sent.*



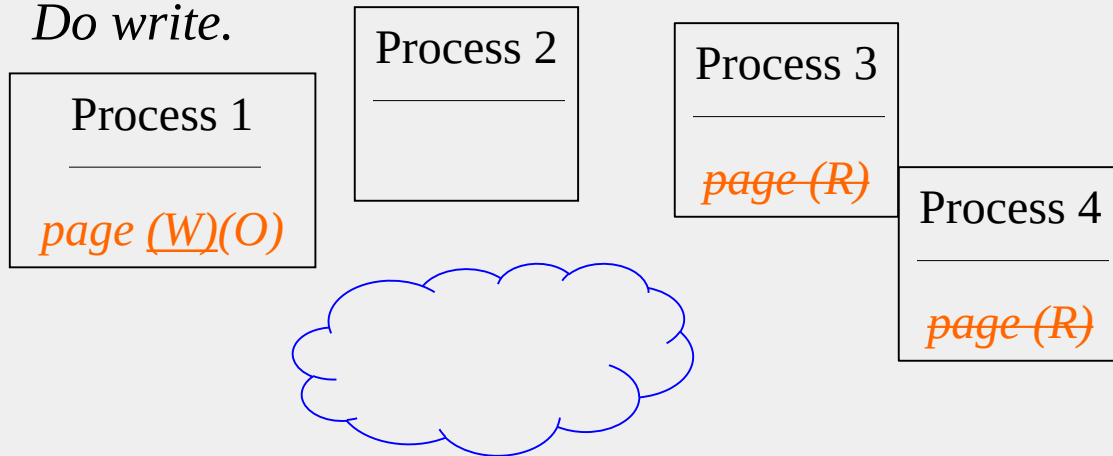
Process 1 Attempting a Write: Scenario 2

- Process 1 is owner (O) has page in R state
- Other processes may also have page in R state
- *Ask other processes to invalidate their copies of page. Use multicast.*
- *Mark page as (W).*
- *Do write.*



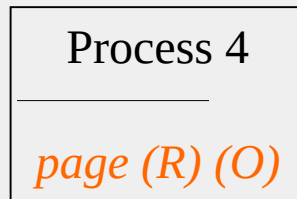
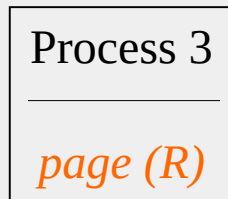
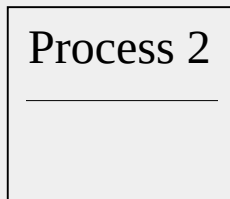
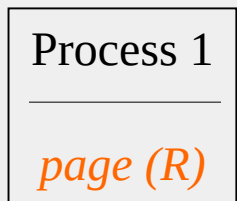
End State: Write Scenario 2

- Process 1 is owner (O) has page in R state
- Other processes may also have page in R state
- *Ask other processes to invalidate their copies of page. Use multicast.*
- *Mark page as (W).*
- *Do write.*



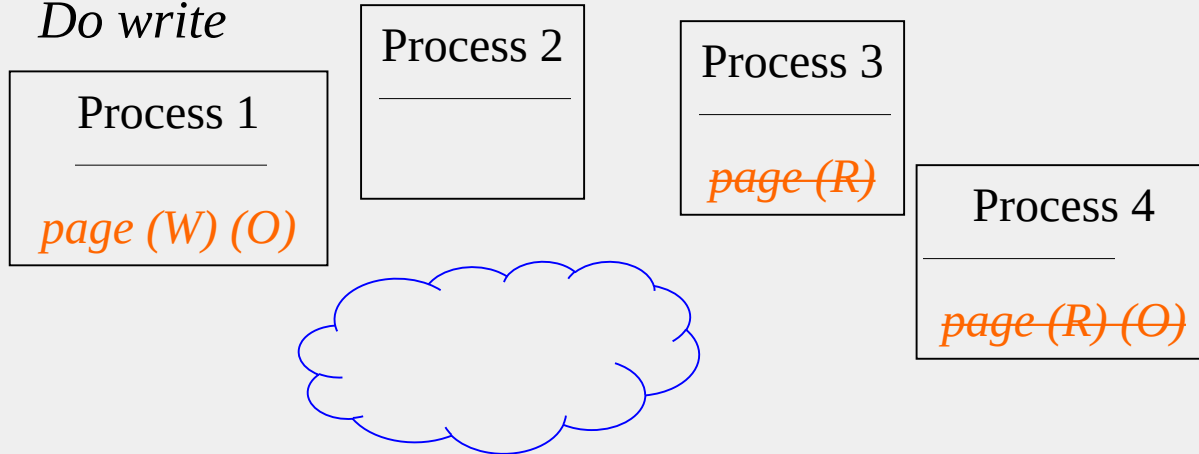
Process 1 Attempting a Write: Scenario 3

- Process 1 has page in R state
- Other processes may also have page in R state, and someone else is owner
- *Ask other processes to invalidate their copies of page. Use multicast.*
- *Mark page as (W), become owner*
- *Do write*



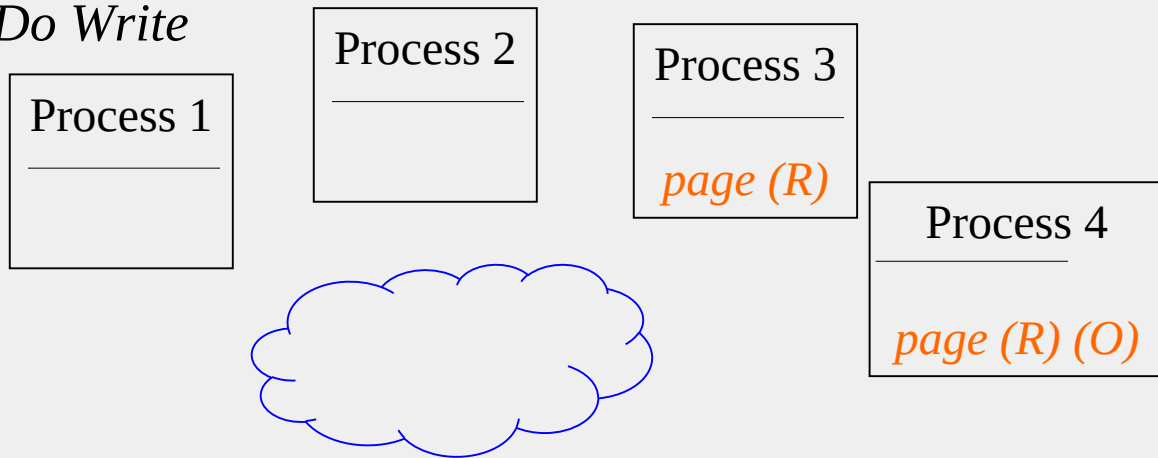
End State: Write Scenario 3

- Process 1 has page in R state
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- *Ask other processes to invalidate their copies of page. Use multicast.*
- *Mark page as (W), become owner*
- *Do write*



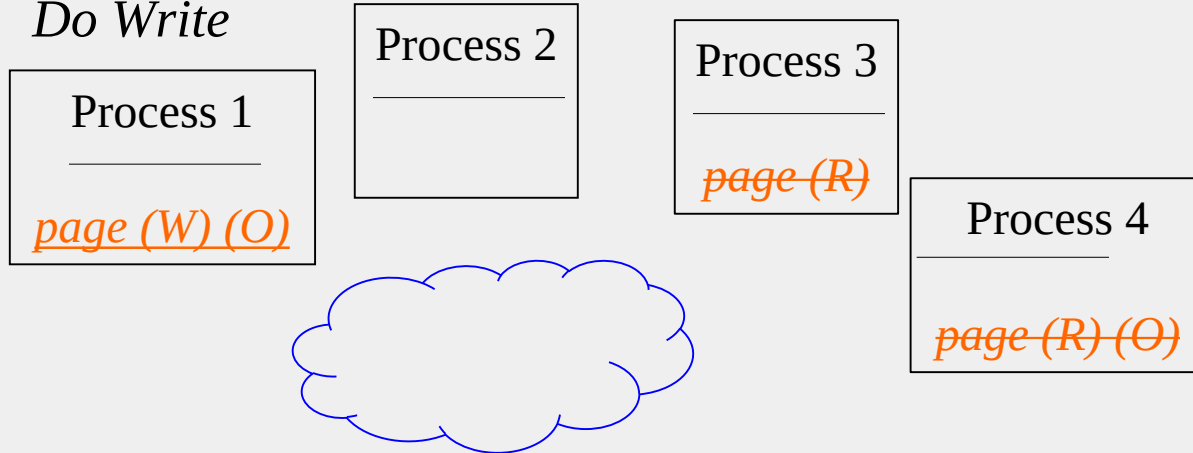
Process 1 Attempting a Write: Scenario 4

- Process 1 does not have page
- Other process(es) has/have page in (R) or (W) state
- *Ask other processes to invalidate their copies of the page. Use multicast.*
- *Fetch all copies; use the latest copy; mark it as (W); become owner*
- *Do Write*



End State: Write Scenario 4

- Process 1 does not have page
- Other process(es) has/have page in (R) or (W) state
- *Ask other processes to invalidate their copies of the page. Use multicast.*
- *Fetch all copies; use the latest copy; mark it as (W); become owner*
- *Do Write*



Invalidate Downsides

- That was the invalidate approach
- If two processes write same page concurrently
 - Flip-flopping behavior where one process invalidates the other
 - Lots of network transfer
 - Can happen when unrelated variables fall on same page
 - Called **false sharing**
- Need to set page size to capture a process' *locality of interest*
- If page size much larger, then have false sharing
- If page size much smaller, then too many page transfers => also inefficient

An Alternative Approach: Update

- Instead: could use **Update** approach
 - Multiple processes allowed to have page in W state
 - On a write to a page, multicast newly written value (or part of page) to all other holders of that page
 - Other processes can then continue reading and writing page
- Update preferable over Invalidate
 - When lots of sharing among processes
 - Writes are to small variables
 - Page sizes large
- Generally though, Invalidate better and preferred option

Consistency

- Whenever multiple processes share data, consistency comes into picture
- DSM systems can be implemented with:
 - Linearizability
 - Sequential Consistency
 - Causal Consistency
 - Pipelined RAM (FIFO) Consistency
 - Eventual Consistency
 - (Also other models like Release consistency)
 - These should be familiar to you from the course!
- As one goes down this order, speed increases while consistency gets weaker

Consistency(Continued)

- **Linearizability** is the strongest form of consistency. This means that all operations are executed in a way as if executed on a single machine, despite the data being distributed across multiple replicas. As a result, every operation returns an up-to-date value.
- Sequential consistency is a **conservative memory model that does not allow any instruction reordering on each core**. This prevents many optimizations and degrades performance. However, not all memory instructions on a single core need to preserve their program order.
- Causal consistency **captures the potential causal relationships between operations, and guarantees that all processes observe causally-related operations in a common order**. In other words, all processes in the system agree on the order of the causally-related operations. A system is **causally consistent** if all agents see (only) causally related events in the same order. This is weaker than sequential consistency.
- PRAM consistency (pipelined random access memory) also known as FIFO consistency. **All processes see memory writes from one process in the order they were issued from the process**. Writes from different processes may be seen in a different order on different processes.
- **Eventual consistency** in general refers to a database's ability to process transactions simultaneously while preserving the integrity of the data.
- Release consistency is one of the synchronization-based [consistency models](#) used in [concurrent programming](#)

Is it Alive?

- DSM was very popular over a decade ago
- But may be making a comeback now
 - Faster networks like Infiniband + SSDs => Remote Direct Memory Access (RDMA) becoming popular
 - Will this grow? Or stay the same as it is right now?
 - Time will tell!

Summary

- DSM = Distributed Shared Memory
 - Processes share pages, rather than sending/receiving messages
 - Useful abstraction: allows processes to use same code as if they were all running over the same OS (multiprocessor OS)
- DSM can be implemented over a message-passing interface
- Invalidate vs. Update protocols